E1Q1.c

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```c
#include <stdio.h>

int main()
{
    int main_arr[7] = {5,7,10,6,3,1,11};
    int a = 6;
    int i = 0;
    int j = 0;
    int temp;
    int bubble_arr[7];
    int insercation_arr[7];
    int selection_arr[7];




    //------Q.1 Part-1 liner search-------
    printf("\n------Q.1 Part-1 liner search-------");
    for(i = 0; i < 7; i++){
        if (a == main_arr[i]){
            printf("\n%d found at %d's place", a, i+1);
            break;
        }
        else
            printf("\n%d's place Checked not found", i+1);
    };


    //---------Q.2 Bubble Sort.-----------
    for(i = 0; i < 7; i++){
        bubble_arr[i] = main_arr[i];
    };

    for(i = 0; i < 7; i++){
        for(j = 0; j < 6 ; j++){
                if (bubble_arr[j]>bubble_arr[j+1]){
                    temp = bubble_arr[j];
                    bubble_arr[j] = bubble_arr[j+1];
                    bubble_arr[j+1] = temp;
                };

        };
    };
    printf("\n---------Q.2 Bubble Sort.-----------");
    for(i = 0; i < 7; i++){
        printf("\n%d",bubble_arr[i]);
    };


    //---------Q.2 Insercation Sort----------
    for(i = 0; i < 7; i++){
        insercation_arr[i] = main_arr[i];
    };

    for(i = 1; i < 7; i++){
```

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```c
    int key = insercation_arr[i];
    j=i-1;
    while(j>=0 && insercation_arr[j]>key){
        insercation_arr[j+1]=insercation_arr[j];
        j=j-1;
    };
    insercation_arr[j+1]=key;
};

printf("\n---------Q.2 Insercation Sort-----------");
for(i = 0; i < 7; i++){
    printf("\n%d",insercation_arr[i]);
};


//---------Q.2 Selection Sort-----------
for(i = 0; i < 7; i++){
    selection_arr[i] = main_arr[i];
};

int temparr[7];
int min_idx;

for (i = 0; i < 6; i++) {
     min_idx = i;

    for (j = i + 1; j < 7; j++) {
        if (selection_arr[j] < selection_arr[min_idx]) {
            min_idx = j;
        };
    };

    temp = selection_arr[min_idx];
    selection_arr[min_idx] = selection_arr[i];
    selection_arr[i] = temp;
};

printf("\n---------Q.2 Selection Sort-----------");
for(i = 0; i < 7; i++){
    printf("\n%d",selection_arr[i]);
};


//----------Q.1 Part-2 Binary Search----------
int st = 0, end = 6;
int z = 0;
printf("\n----------Q.1 Part-2 Binary Search----------");

while (st <= end){
    int mid = (st + end) / 2;
    if(selection_arr[mid] == a){
        printf("\n%d found at %d's place", a, mid+1);
        z=1;
        break;
    }

    else if(selection_arr[mid] > a){
        end = mid-1;
```

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114         }
115 
116         else if(selection_arr[mid] < a){
117             st = mid+1;
118         }
119     }
120 
121     if(z == 0) {
122         printf("\n%d not found", a);
123     }
124 
125 
126 
127     //---------Q3 Max and Min-----------
128     printf("\n---------Q3 Max and Min-----------");
129     printf("\nMin:%d", selection_arr[0]);
130     printf("\nMax:%d", selection_arr[6]);
131     printf("\nSecond Min:%d", selection_arr[1]);
132     printf("\nSecond Max:%d", selection_arr[5]);
133 
134 }
```

Output :

```
------Q.1 Part-1 liner search-------
1's place Checked not found
2's place Checked not found
3's place Checked not found
6 found at 4's place
---------Q.2 Bubble Sort.-----------
1
3
5
6
7
10
11
---------Q.2 Insercation Sort-----------
1
3
5
6
7
10
11
---------Q.2 Selection Sort-----------
1
3
5
6
7
10
11
----------Q.1 Part-2 Binary Search----------
6 found at 4's place
---------Q3 Max and Min-----------
Min:1
Max:11
Second Min:3
Second Max:10
```